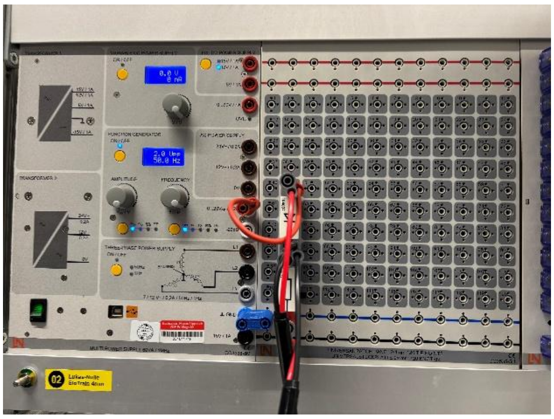
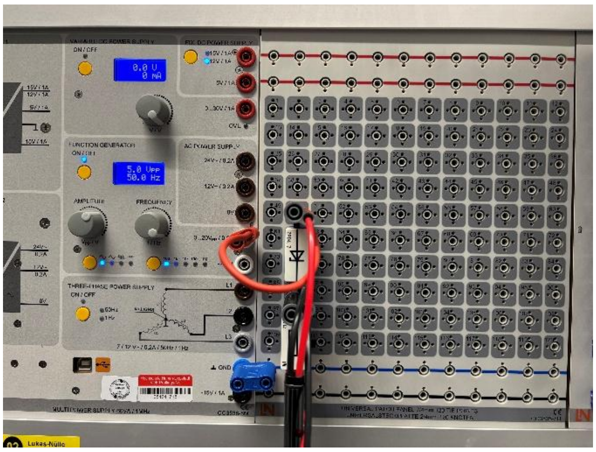
Task-1:



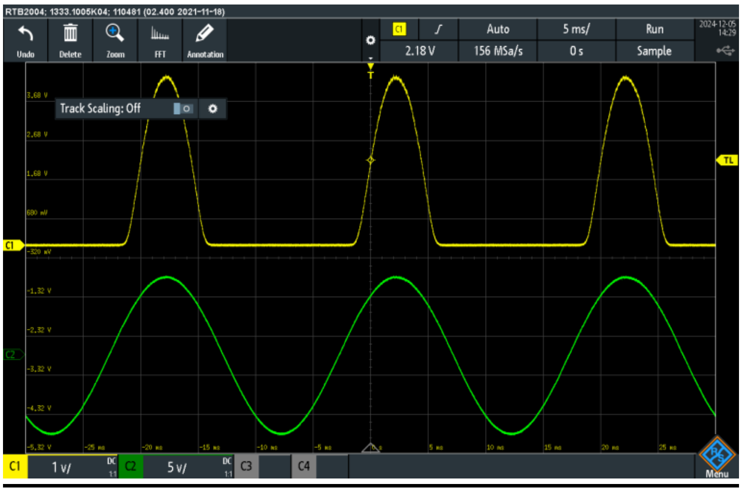
Applying $V_{SS} = i_D * R + V_D$ on the result and it matched.

Task-2:



Here , we see that reverse current $i_D = 0$, and breakdown voltage $V_D = 4.7$ V

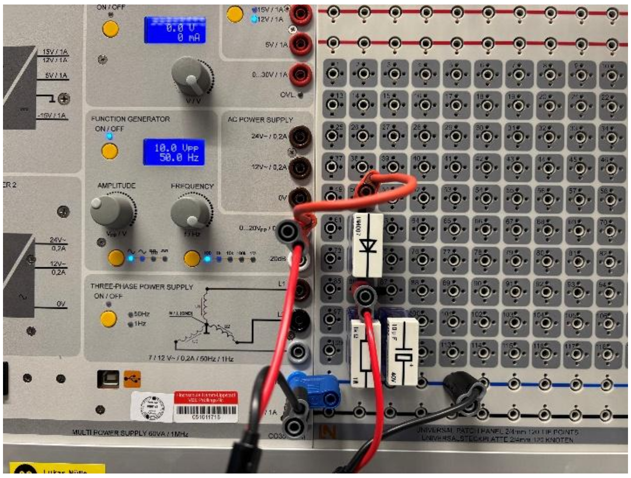
Task-3:



We got;

$V_R = V_S - (V_{D1} + V_{D2}) = 4.7V$

Task-4:



Output Voltage (Rectified Peak): 4.35V

Ripple Voltage (V_{ripple}) = $V_{out(max)}$ – $V_{in(min)}$ = 4.35-0.940=3.4V